

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 – Digital Electronics
Spring Term 2026

Lab. 11-Transmission Gate/Domino Logic

Issued: 4/15/2026

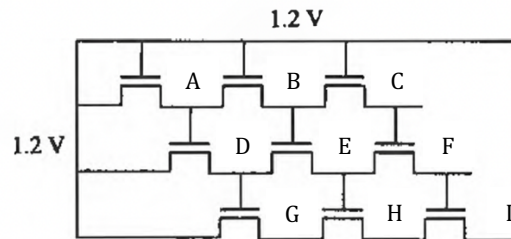
Due: 4/17/2026

Question 1:

Objectives:

- NMOS Pass Transistor Configurations

- (a) Compute all the voltages at all the nodes for the pass transistor circuit of Figure 1 including the body effect. Use $0.13\mu\text{m}$ technology parameters. Assume that all nodes are initially at 0 V.

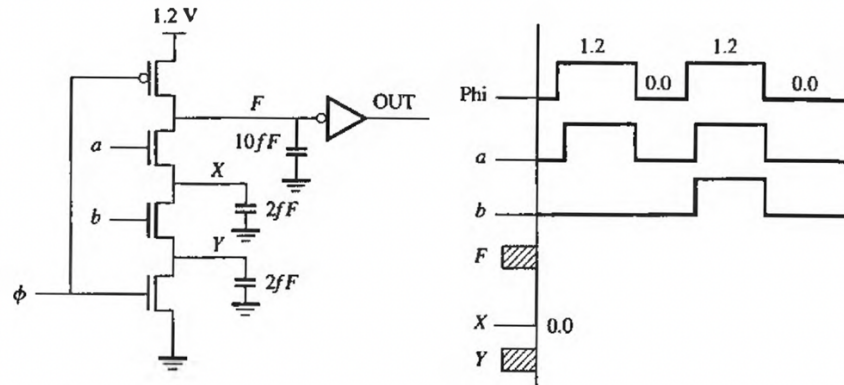


- (b) Verify the results with LTSPICE simulations.

Question 2:

Objectives:

- 2 input AND gate with Domino Logic



i. The precharged circuit of Figure 2 is a 2-input AND gate. Draw the waveforms at nodes F , X , and Y as a function of time. You can assume short rise and fall times (i.e., you do not need to calculate them), but the final value in each phase must be correct. Only the capacitances shown here need to be considered. All devices are minimum size. Both F and Y start out unknown. Use 130nm parameters.

ii. Next, Build the circuit shown in LTSPICE and run simulations to verify the waveforms obtained in (i).

Report Submission:

Submit your work on LMS under **Labs**

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm.